

Project Report

Honeypot Attack Detection using Cowrie and Splunk

1. Executive Summary

This project demonstrates the deployment of a Cowrie honeypot to capture malicious SSH/Telnet activity, with logs forwarded to Splunk Enterprise for centralized monitoring. The setup simulates a real SOC workflow: attacker reconnaissance → honeypot capture → log forwarding → SIEM analysis.

Key Skills Highlighted: Network security, SIEM (Splunk), log forwarding, honeypot deployment, Linux/Windows integration, attacker simulation.

2. System Architecture

Flow:

Kali (Attacker) → Ubuntu (Cowrie Honeypot) → Splunk Forwarder → Splunk Enterprise (Windows)

- **Kali Linux:** Simulated attacker (Nmap scans).
- **Cowrie Honeypot (Ubuntu):** Captures scan attempts and logs them.
- **Splunk Universal Forwarder:** Ships logs securely to Splunk Enterprise.
- **Splunk Enterprise (Windows):** Centralized log collection and analysis.

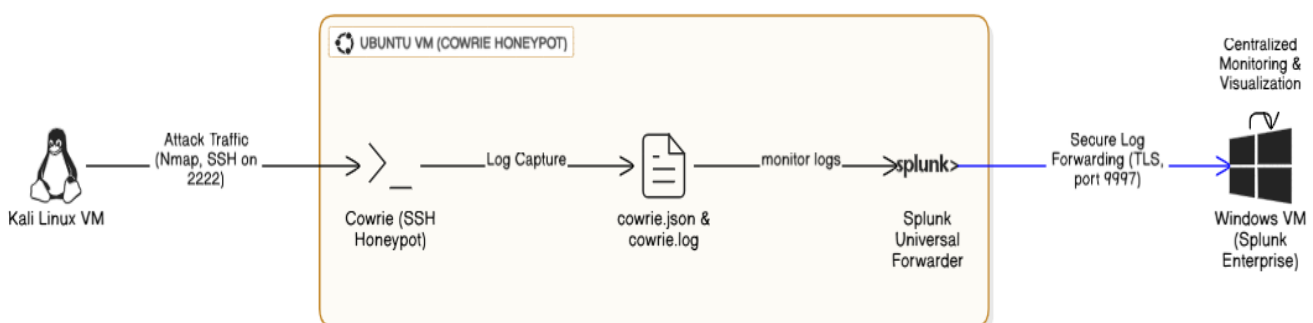


Fig 1: -Architecture Diagram

3. Implementation

3.1 Environment Setup:

- VirtualBox with three VMs: Ubuntu (honeypot), Kali (attacker), Windows (Splunk).
- Configured host-only networking for controlled attack simulation.

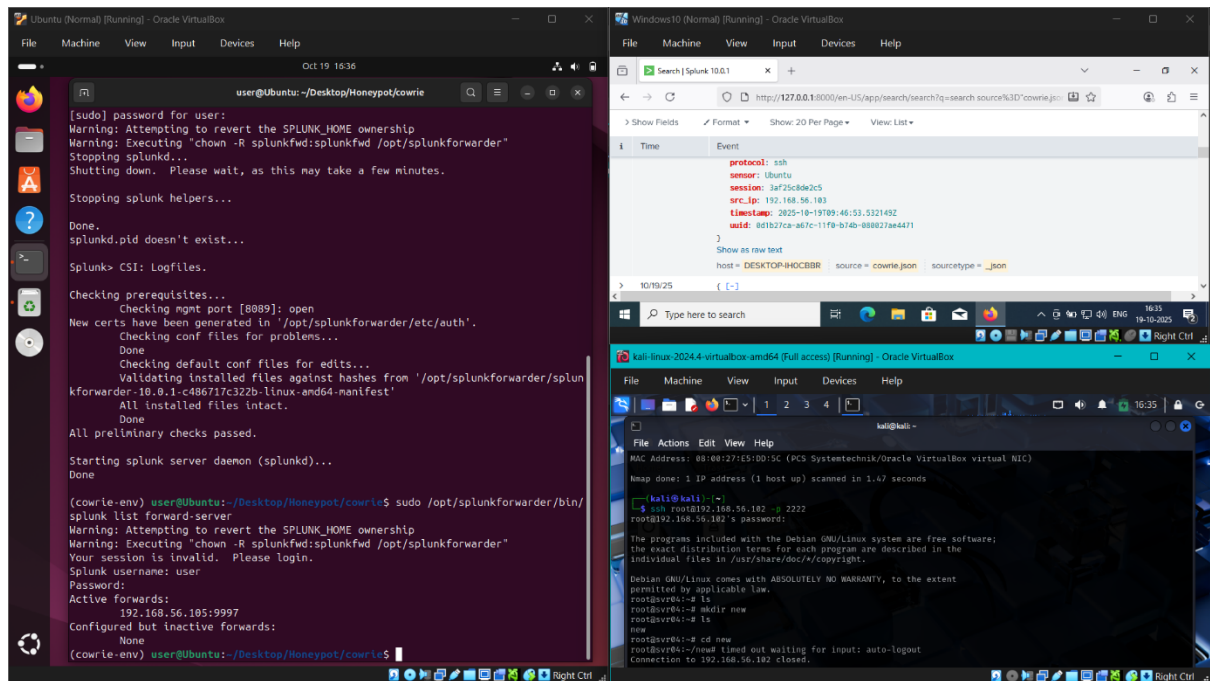


Fig 2: -Kali, Ubuntu & Windows VMs

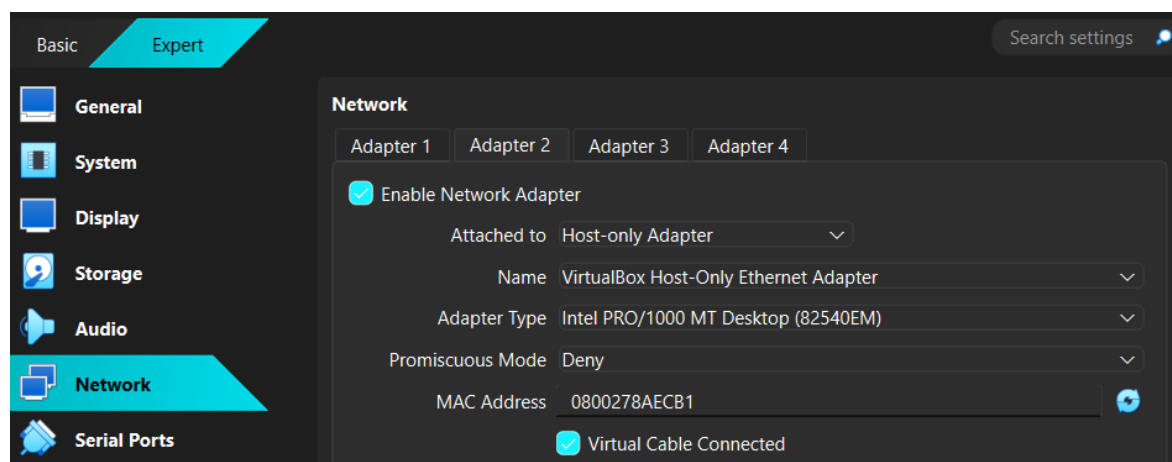


Fig 3: - VirtualBox VM setup

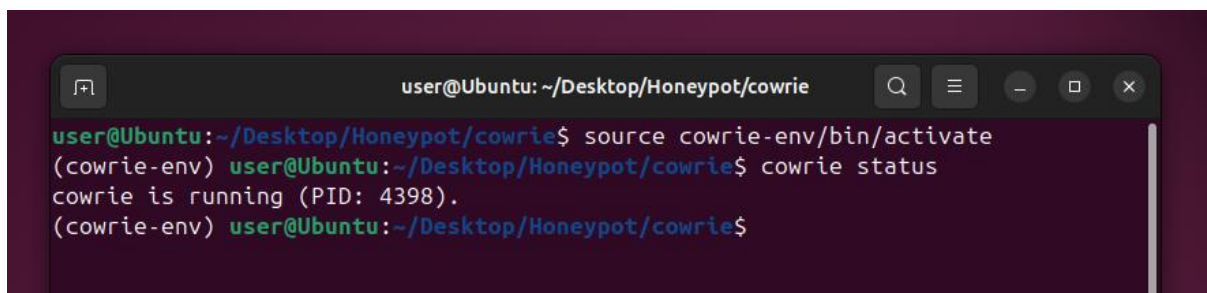
3.2 Cowrie Deployment:

- Installed Cowrie in a Python virtual environment (cowrie-env).
- Activated environment and started honeypot:

```
source cowrie-env/bin/activate  
bin/cowrie start
```

- Logs generated in

```
home/user/Desktop/Honeypot/cowrie/var/log/cowrie/  
(cowrie.json, cowrie.log)
```

A terminal window titled 'user@Ubuntu: ~/Desktop/Honeypot/cowrie' showing the execution of 'source cowrie-env/bin/activate' and 'cowrie status'. The output of 'cowrie status' is 'cowrie is running (PID: 4398)'.

```
user@Ubuntu: ~/Desktop/Honeypot/cowrie  
user@Ubuntu:~/Desktop/Honeypot/cowrie$ source cowrie-env/bin/activate  
(cowrie-env) user@Ubuntu:~/Desktop/Honeypot/cowrie$ cowrie status  
cowrie is running (PID: 4398).  
(cowrie-env) user@Ubuntu:~/Desktop/Honeypot/cowrie$
```

Fig 4: -Cowrie start confirmation

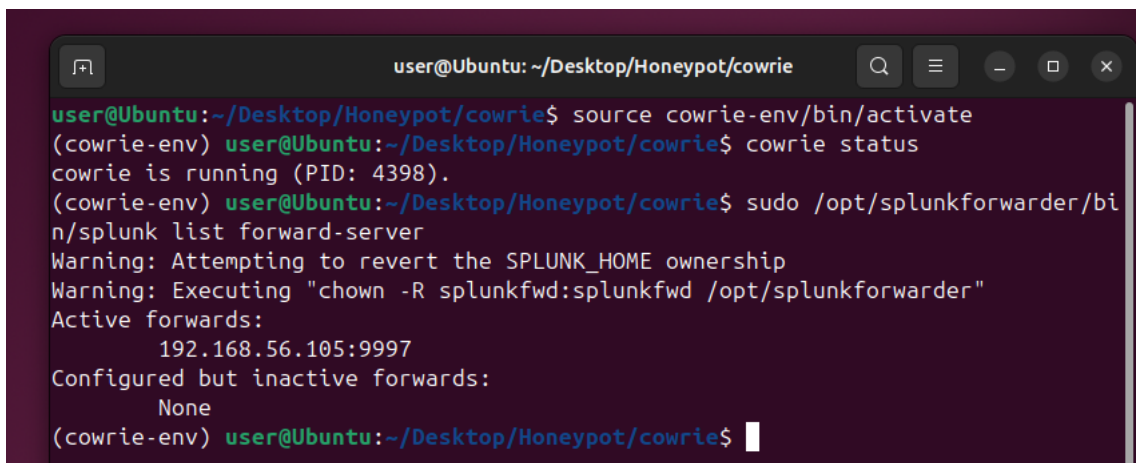
3.3 Splunk Forwarder Configuration:

- Installed Splunk Universal Forwarder on Ubuntu.
- Added monitor for Cowrie logs:

```
sudo /opt/splunkforwarder/bin/splunk add monitor  
/home/user/Desktop/Honeypot/cowrie/var/log/cowrie/cowrie.json
```

- Verified with:

```
sudo /opt/splunkforwarder/bin/splunk list monitor
```

A terminal window titled 'user@Ubuntu: ~/Desktop/Honeypot/cowrie' showing the execution of 'source cowrie-env/bin/activate', 'cowrie status', and 'sudo /opt/splunkforwarder/bin/splunk list forward-server'. The output of 'splunk list forward-server' shows 'Active forwards: 192.168.56.105:9997' and 'Configured but inactive forwards: None'.

```
user@Ubuntu: ~/Desktop/Honeypot/cowrie  
user@Ubuntu:~/Desktop/Honeypot/cowrie$ source cowrie-env/bin/activate  
(cowrie-env) user@Ubuntu:~/Desktop/Honeypot/cowrie$ cowrie status  
cowrie is running (PID: 4398).  
(cowrie-env) user@Ubuntu:~/Desktop/Honeypot/cowrie$ sudo /opt/splunkforwarder/bin/splunk list forward-server  
Warning: Attempting to revert the SPLUNK_HOME ownership  
Warning: Executing "chown -R splunkfwd:splunkfwd /opt/splunkforwarder"  
Active forwards:  
192.168.56.105:9997  
Configured but inactive forwards:  
None  
(cowrie-env) user@Ubuntu:~/Desktop/Honeypot/cowrie$
```

Fig 5: -Splunk forwarder monitor list

3.4 Attack Simulation:

- From Kali, ran reconnaissance scans against the honeypot:

```
nmap 192.168.56.102
```

- Attempted to connect to the honeypot using SSH:

```
ssh root@192.168.56.102 -p 2222
```

- Cowrie captured the SSH session attempt, logging details such as:
 - Source IP (192.168.56.10)
 - Username used (root)
 - Connection attempt metadata

```
(kali@kali)-[~]
$ nmap 192.168.56.102
Starting Nmap 7.95 ( https://nmap.org ) at 2025-10-19 15:26 IST
Nmap scan report for 192.168.56.102
Host is up (0.0017s latency).
Not shown: 999 closed tcp ports (reset)
PORT      STATE SERVICE
2222/tcp  open  EtherNetIP-1
MAC Address: 08:00:27:E5:DD:5C (PCS Systemtechnik/Oracle VirtualBox virtual NIC)

Nmap done: 1 IP address (1 host up) scanned in 1.66 seconds

(kali@kali)-[~]
$ ssh root@192.168.56.102 -p 2222
root@192.168.56.102's password:

The programs included with the Debian GNU/Linux system are free software;
the exact distribution terms for each program are described in the
individual files in /usr/share/doc/*/copyright.

Debian GNU/Linux comes with ABSOLUTELY NO WARRANTY, to the extent
permitted by applicable law.
root@svr04:~# ls
root@svr04:~# mkdir new
root@svr04:~# ls
new
root@svr04:~# cd new
root@svr04:~/new#
```

Fig 6: -Kali Nmap scan and SSH login attempt

```
2025-10-19T09:46:23.186209Z [HoneyPotSSHTransport,0,192.168.56.103] login attempt [b'root'/b'pass'] succeeded
2025-10-19T09:46:23.186480Z [HoneyPotSSHTransport,0,192.168.56.103] Initialized emulated server as architecture: linux-x64-lsb
2025-10-19T09:46:23.186662Z [cowrie.ssh.userauth.HoneyPotSSHUserAuthServer#debug] b'root' authenticated with b'password'
2025-10-19T09:46:23.186841Z [cowrie.ssh.transport.HoneyPotSSHTransport#debug] starting service b'ssh-connection'
2025-10-19T09:46:23.188728Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got channel b'session' request
2025-10-19T09:46:23.188961Z [cowrie.ssh.session.HoneyPotSSHSession#info] channel open
2025-10-19T09:46:23.189030Z [cowrie.ssh.connection.CowrieSSHConnection#debug] got global b'no-more-sessions@openssh.com' request
2025-10-19T09:46:23.219394Z [twisted.conch.ssh.session#info] Handling pty request: b'xterm-256color' (55, 153, 0, 0)
2025-10-19T09:46:23.219517Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,192.168.56.103] Terminal Size: 153 55
2025-10-19T09:46:23.219972Z [SSHChannel session (0) on SSHService b'ssh-connection' on HoneyPotSSHTransport,0,192.168.56.103] request_env: LANG=en_US.UTF-8
2025-10-19T09:46:23.220304Z [twisted.conch.ssh.session#info] Getting shell
2025-10-19T09:46:38.260796Z [HoneyPotSSHTransport,0,192.168.56.103] CMD: ls
2025-10-19T09:46:38.261138Z [HoneyPotSSHTransport,0,192.168.56.103] Command found: ls
2025-10-19T09:46:45.040581Z [HoneyPotSSHTransport,0,192.168.56.103] CMD: mkdir new
2025-10-19T09:46:45.040916Z [HoneyPotSSHTransport,0,192.168.56.103] Command found: mkdir new
2025-10-19T09:46:47.784536Z [HoneyPotSSHTransport,0,192.168.56.103] CMD: ls
2025-10-19T09:46:47.784863Z [HoneyPotSSHTransport,0,192.168.56.103] Command found: ls
2025-10-19T09:46:53.532149Z [HoneyPotSSHTransport,0,192.168.56.103] CMD: cd new
2025-10-19T09:46:53.532902Z [HoneyPotSSHTransport,0,192.168.56.103] Command found: cd new
2025-10-19T09:49:52.914295Z [twisted.conch.ssh.session#info] exitCode: 1
2025-10-19T09:49:52.915528Z [cowrie.ssh.connection.CowrieSSHConnection#debug] sending request b'exit-status'
2025-10-19T09:49:52.915528Z [-] Closing TTY Log: var/lib/cowrie/tty/748533748bd6e980eccc8035c7441341ab9bbfae4e4ade191f65c2fa1ca264a after 209.7 seconds
```

Fig 7: -Cowrie log entries

4. Results

- **Cowrie Logs:** Captured both the Nmap scan probes and the SSH connection attempt.
- **Splunk Enterprise:** Successfully ingested these logs, making them searchable and visualizable.
- **Insights:**
 - Reconnaissance (Nmap) and access attempts (SSH) were both detected.
 - Demonstrated how honeypots can log attacker behavior beyond simple scans.

← → ↻ http://127.0.0.1:8000/en-US/app/search/search?q=search source%3D"cowrie.json" host%3D"DESKTOP-IH0CBBR" sou

Events (12) Patterns Statistics Visualization

Timeline format Zoom Out Zoom to Selection Deselect

Format Show: 20 Per Page View: List

< Hide Fields All Fields

SELECTED FIELDS

- host 1
- source 1
- sourcetype 1

INTERESTING FIELDS

- eventid 9
- index 1
- input 3
- linecount 1
- message 10
- protocol 1
- punct 10
- sensor 1
- session 1
- splunk_server 1
- src_ip 1
- timestamp 13
- uuid 1

20 more fields

+ Extract New Fields

i	Time	Event
>	10/19/25 4:00:37.000 AM	{ [-] duration: 224.6 eventid: cowrie.session.closed message: Connection lost after 224.6 seconds protocol: ssh sensor: Ubuntu session: 3af25c8de2c5 src_ip: 192.168.56.103 timestamp: 2025-10-19T09:49:52.922793Z uuid: 0d1b27ca-a67c-11f0-b74b-080027ae4471 } Show as raw text host = DESKTOP-IH0CBBR source = cowrie.json sourcetype = _json
>	10/19/25 4:00:37.000 AM	{ [-] eventid: cowrie.command.input input: cd new message: CMD: cd new protocol: ssh sensor: Ubuntu session: 3af25c8de2c5 src_ip: 192.168.56.103 timestamp: 2025-10-19T09:46:53.532149Z uuid: 0d1b27ca-a67c-11f0-b74b-080027ae4471 } Show as raw text host = DESKTOP-IH0CBBR source = cowrie.json sourcetype = _json
>	10/19/25 4:00:37.000 AM	{ [-] eventid: cowrie.command.input

Fig 8: -Splunk search results showing Cowrie logs

5. Key Takeaways

- Honeypots provide early warning of reconnaissance and access attempts.
- Splunk integration enables centralized monitoring and scalable analysis.
- Project showcases hands-on SOC skills: log collection, parsing, and attacker behavior analysis.

6. Future Enhancements

- Deploy multiple honeypots across different networks.
- Add automated alerting in Splunk for unusual scan or login activity.
- Compare with ELK/Wazuh for open-source SIEM alternatives.

7. Career Relevance

This project demonstrates practical expertise in:

- **Cybersecurity operations** (honeypot deployment, attacker simulation).
- **SIEM tools** (Splunk configuration, log forwarding, event analysis).
- **Networking & Linux administration** (VM setup, SSH, Nmap).

It positions me for roles such as SOC Analyst, Security Engineer, or Network Security Specialist.

8. References

1. Cowrie Honeypot Documentation. *Cowrie Project*. Available at: <https://docs.cowrie.org/>
2. Splunk Documentation. *Splunk Enterprise and Universal Forwarder Guides*. Available at: <https://docs.splunk.com/>
3. Nmap Reference Guide. *Nmap Security Scanner Project*. Available at: <https://nmap.org/book/man.html>
4. VirtualBox User Manual. *Oracle VM VirtualBox*. Available at: <https://www.virtualbox.org/manual/>